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Jeff Shen

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Education

Ph.D., Princeton University Astrophysical Sciences. 2022 - Present
H.B.Sc., University of Toronto. Astronomy & Astrophysics and Statistics. 2018-2022

Publications

3. **Jeff Shen**, Joshua S. Speagle, Neige Frankel, J. Ted Mackereth, Yuan-Sen Ting, Jo Bovy. Disentangling Stellar Age Estimates from Galactic Chemodynamical Evolution, in prep.

2. **Jeff Shen**, Gwendolyn M. Eadie, Norman Murray, Dennis Zaritsky, Joshua S. Speagle, Yuan-Sen Ting, Charlie Conroy, Phillip A. Cargile, Benjamin D. Johnson, Rohan P. Naidu, Jiwon Jesse Han. The Mass of the Milky Way from the H3 Survey. ApJ, 925, 1. 2111.09327.

1. **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$. 2021, ApJ, 917, 79. 2105.11572.

Research Experience

2021 Sep - Present **Narrowband upgrade for the Dragonfly Telephoto Array**
Supervisor: Prof. Roberto Abraham
Senior thesis. Containerized camera control software in C++ and Rust with real-time, multi-threaded data processing.

2021 Jul - Present **Constraining the strength of thermal tides with Stan**
Supervisor: Prof. Norman Murray
Independent research. Fitting dynamical models for the angular momentum of the Earth-Moon system to constrain the strength of thermal tides over geological time.

2021 May - Present **Predicting the ages of stars with machine learning**
Supervisors: Dr. Joshua Speagle, Dr. Neige Frankel, Dr. J. Ted Mackereth
Independent research. Probabilistic predictions of stellar ages using normalizing flows. Quantifying the contribution of stellar/Galactic information to age estimates.

2020 Aug - 2021 Nov **Estimating the mass of the Galaxy with Bayesian hierarchical models**
Supervisors: Prof. Gwendolyn Eadie, Prof. Norman Murray, Dennis Zaritsky, Dr. Joshua Speagle
Undergraduate fellowship with the Canadian Institute for Theoretical Astrophysics (CITA). Developed a hierarchical Bayesian model for modelling the mass of the Galaxy using a fast and scalable sampling algorithm. Collaborated with the H3 Survey team and presented results in ApJ.

2020 May - 2021 May **Molecular gas in high-redshift galaxies with ALMA**
Supervisor: Prof. Allison Man
Summer undergraduate research program with the Dunlap Institute. Analyzed ALMA data to infer molecular gas masses in several gravitationally lensed high-redshift galaxies. Presented results in a first-author publication in ApJ.

Honours and Awards

2022 Jan **Astrostatistics Student Paper Competition Finalist**
Astrostatistics Interest Group of the American Statistical Association

2021 - 2022	**Dean's List Scholar** University of Toronto
2021 May - 2021 Aug	**Undergraduate Summer Research Fellowship, ($8,396 CAD)** Canadian Institute for Theoretical Astrophysics, University of Toronto
2020 May - 2020 Aug	**Summer Undergraduate Research Program Award ($9,500 CAD)** Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Talks

2021 August	**SDSS 2021 Collaboration Meeting, JHU/Online** Lightning talk: "Predicting the ages of stars with machine learning."
2021 May	**Stellar Stats Workshop, UofT/Online** Lightning talk: "Estimating the mass distribution of the Milky Way with Bayesian multilevel models."
2021 May	**Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online** Lightning talk: "Molecular gas in a gravitationally lensed galaxy group at z = 2.9."
2021 Apr	**Vancouver Area Cosmology Meeting, UBC/SFU/TRIUMF/Online** "Molecular gas in gravitationally lensed galaxies at z = 2.9."
2020 Oct	**REQUIEM Galaxies Telecon, Online** "Molecular gas in gravitationally lensed galaxies at z = 2.9."

Posters

2021 May	**Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online** "Molecular gas in a gravitationally lensed galaxy group at z = 2.9."
2020 Aug	**Summer Undergraduate Research Program Poster Session, UofT/Online** "Molecular gas in a gravitationally lensed galaxy group at z = 2.9." Awarded prize for one of top three posters.

Outreach

2019 - 2021	**Space Science Journalist (Current in Space segment)**, *The Star Spot* podcast Curate, write and record stories about the latest research and news in astronomy. Episodes 177 - end of show.
2015 - 2018	**Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C. Coordinated and delivered workshops to prepare students for national programming contest.

Media

2021 Jul	**National Ratio Astronomy Organization (NRAO) eNews** Digital newsletter feature: "Molecular gas in high redshift galaxies". `https://science.nrao.edu/enews/14.7/index.shtml#molecular_gas`
2020 Jul	**SURP Student of the Week Feature** Interview: `https://www.dunlap.utoronto.ca/surp-sow/sow12020/`

Technical Skills

JAX, Stan, Pytorch, SQL, R, C++, Rust, BLAS/LAPACK, LaTeX, regex, bash, git, Docker, Javascript, React